

SAMRUDDHI KALEDESHMUKH

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SUMMARY

MS in Data Science student driven by a deep curiosity for building deep learning systems from first principles rather than treating them as black boxes — from Vision/Swin Transformers to RL-based detection pipelines built entirely from scratch. My research on mHealth for dementia care fuels my belief that AI should be built around real human needs. I hope to eventually pursue a PhD, and am using my master's to dig deeper into open research questions.

EDUCATION

Northeastern University, San Jose, CA

Sept 2025 – May 2027

Master of Science in Data Science

Relevant Courses: Algorithms, Machine Learning, Essentials of Data Science, Data Mining Techniques, Cloud Computing

Cummins College of Engineering for Women, Pune, India

Aug 2020 – May 2025

Bachelor of Technology, Computer Science

Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Soft Computing, Digital Image Processing

EXPERIENCE

Concept Information Technology Pvt. Ltd., Pune, India

Summer 2022

Software Engineering Intern

- Developed and optimized backend and frontend features across internal applications, improving page load times and system responsiveness by 30%.
- Built and integrated RESTful APIs to support data exchange between services, reducing manual processing and improving data consistency.
- Automated routine workflows and scripts, cutting repetitive engineering effort by 25%, following Git-based version control best practices.

Cummins College of Engineering for Women, Pune, India

2024 – 2025

Research Assistant — Advisor: Prof. Dr. Chhaya Santosh Gosavi

- Co-authored a comprehensive literature review on mHealth technologies for dementia care and caregiver support, evaluating 20+ existing applications across Android and iOS platforms.
- Synthesized findings across usability, adoption, and clinical validation to propose a holistic mobile solution integrating reminders, cognitive games, GPS-based safety alerts, and a voice-to-text diary.

Research Assistant — Research Guide: Prof. Dr. Anjali Naik

- Formulated the object detection problem (bounding-box localization) as a dynamic Markov Decision Process (MDP) using deep reinforcement learning.
- Trained and evaluated the approach on the Pascal VOC2012 dataset, contributing to a publication on object detection using deep reinforcement learning.

RESEARCH PROJECTS

Building a Vision Transformer from Scratch for Remote Sensing Change Detection

Spring 2026

Computer Vision · Vision Transformers · Swin Transformers · PyTorch · Remote Sensing

- Implemented a Siamese Vision Transformer from scratch in PyTorch (custom patch embedding, 12-layer multi-head self-attention, FPN-style decoder) and extended it with a Siamese Swin Transformer using hierarchical shifted-window attention for pixel-level change detection on bi-temporal satellite imagery.
- Trained and evaluated both models on the LEVIR-CD benchmark (7,120 training pairs); the Swin variant achieved an F1 score of **0.8613** and IoU of **0.7564**, a **+3.77% F1** and **+5.63% IoU** gain over the ViT baseline.
- Applied multi-crop augmentation and cosine LR scheduling with warm restarts to stabilize training and improve generalization on a highly imbalanced (~2–10% changed pixels) dataset.
- Independently presented this work as a poster at the Research and Innovation Showcase 2026.

DEMCARE: Holistic Mobile Application for Dementia Patients

Pune, April 2025

Mobile Health · Reinforcement Learning · Adaptive Systems · Applied ML

- Generated ML-ready data pipelines processing 10K+ daily behavioral events, enabling 95% signal consistency for downstream modeling and analytics.
- Implemented adaptive cognitive games, GPS-based safety alerts, and a voice-to-text diary, driving 75% higher user engagement and 80% faster real-time risk detection.

Low-Resource Medical Imaging via Transfer Learning

with Aayush Katoch

Transfer Learning · Limited Data · Uncertainty · Statistical Evaluation

- Co-developed a study on model generalization under extreme data scarcity (~300 samples), benchmarking three transfer learning paradigms (scratch, ImageNet, medical-domain pretraining) with k-fold cross-validation; achieved **92.02% accuracy** and **0.9875 AUC** (14% gain), producing interpretable findings on when domain-adjacent pretraining generalizes reliably.

TECHNICAL SKILLS

Research Practices: Literature Review, Hypothesis-Driven Experimentation, Model Evaluation & Benchmarking, Data Pipeline Design, Reproducible Training Workflows

Languages: Java, Python, SQL, C, C++

Frameworks & Libraries: Keras, TensorFlow, PyTorch, Scikit-learn, OpenVINO, Torch, React, Next.js, FastAPI, Flask, Node.js, Pandas, NumPy

AI/ML & NLP: LLM Integration, RAG Systems, Semantic Search, NLP Processing, Prompt Engineering

Deep Learning: CNNs, RNNs, Vision Transformers, Swin Transformers, GANs, VAEs

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Cloud & DevOps: AWS, Google Cloud, Docker, Kubernetes, Git, GitLab

Web Technologies: REST APIs, OAuth Integration, WebSockets, State Management, API Integration